BRAC UNIVERSITY
Inspiring Excellence

Linear Algebra & Fourier Analysis

Quiz-03

January 7, 2026

Total - 20 Marks

(You need to answer **ANY TWO** questions)

Name:

ID:

Section:

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1. (a)
$$2x + 4y = 6$$
$$x + 2y = 3$$
Give reasons based on determinants.

   (b) Solve the system using $X = A^{-1}B$:
$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}.$$

2. (a) A team of four has to be selected from 6 boys and 4 girls. How many different ways can a team be selected if at least one boy must be in the team?

   (b) How many triangles can be formed using 10 points in a plane, out of which 4 are collinear?

   (c) How many parallelograms are formed when a set of 5 parallel lines intersects a set of 4 parallel lines?

   (d) Evaluate: $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$.

3. Find k so that f is continuous at **both** $x = 0$ and $x = 2$:
$$f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ k+4, & x = 2 \end{cases}$$
And $f(0) = k - 2$.

Best of Luck!